# EE5239 Introduction to Nonlinear Optimization

# Lecture 6: Optimization Over Constraints

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# Summary

- When we are no longer "free" to explore the entire space, many interesting things happen

  1. Optimality condition needs to be updated
  2. Algorithm needs to be updated
  3. But surprisingly, most theory still works!

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- After this lecture, I hope you will be able to
  1. Understand the geometric view of optimality condition
  2. Understand how to deal with "simplest" constraints
  3. Know how to perform an important operation called "projection"
- **Readings**
  1. Section 2.1.1 (version 2)
  2. Section 3.1.1 (version 3)

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- Please carefully read the notes accompanying HW 3.

# Example: Support Vector Machine

- **Training data set**: $\{\mathbf{a}_i, b_i\}_{i=1}^{M}$, $\mathbf{a}_i \in \mathbb{R}^K$, $b_i \in \{-1,\ 1\}$
- **Objective**: learn the classification function $f(\tilde{\mathbf{a}}; \tilde{\mathbf{x}}) = \mathsf{sgn}(\tilde{\mathbf{a}}^T \tilde{\mathbf{x}})$
- Let $\tilde{\mathbf{a}} = [\mathbf{a}; 1]$, $\tilde{\mathbf{x}} = [\mathbf{x}; x_0]$
- Suppose the training data are linearly separable
- Select two hyperplanes to separate the data points, and then try to maximize their distance

# Example: Support Vector Machine

- **Formulation**: Consider the following problem [see reading material]

$$\begin{aligned} \min_{\tilde{\mathbf{x}}} \quad & \tilde{\mathbf{x}}^T \tilde{\mathbf{x}} \\ \text{s.t.} \quad & b_i(\tilde{\mathbf{a}}_i^T \tilde{\mathbf{x}}) \ge 1, \ \forall\ i \end{aligned} \tag{0.1}$$

- The objective minimizes the inverse of the distance (or equivalently, maximizes the distance) between two supporting planes
- The blue part says there is no classification error
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- **Intuition**: Find the separating plane far away from both classes
- **Question**: How to characterize the optimal solution? Have a feasible solution? Which algorithm?

# Constrained Optimization Problem

$$\boxed{\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & x \in X, \end{array}}$$

- Here $f$ is continuously differentiable, $X$ is a convex set

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- Convex set $X$ means we allow the following types of constraints
  - $g(x) \leq 0$ where $g(x)$ is a convex function
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- Why $g(x) = 0$ is not a convex set? Consider $x_1^2 + x_2^2 = 1$

# Optimality Conditions

$$\boxed{\begin{aligned} &\text{minimize} \quad f(x) \\ &\text{subject to} \quad x \in X, \end{aligned}}$$

- **Question:** How to characterize the global/local optimal solution $x^*$?

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- Suppose $X$ is a convex set; $f$ is continuously differentiable
- **Claim (a)** If $x^*$ is a local minimum of the above constrained minimization problem, then (necessary condition)

$$\langle \nabla f(x^*), x - x^* \rangle \geq 0, \; \forall \; x \in X$$

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- **Claim (b)** If further assume that, $f$ is a convex function, this condition is also sufficient for $x^*$ to minimize $f$ over $X$
- **Note:** If $f$ is nonconvex, solutions satisfying this condition is called stationary point (nowhere to move)

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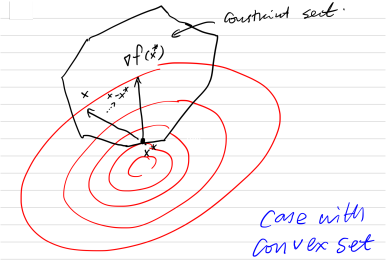


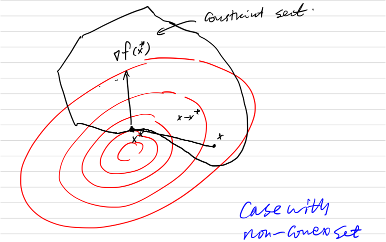


**Note:** It is important to realize the difference between having convex constraint set $X$, and non-convex constraint set $X$

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- Think about a simple 1-dimensional problem

$$\min\ x^2, \quad \text{s.t.} \quad 3 \le x \le 5 \tag{0.2}$$

# Proof of Claim a

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- By the MVT, for every $\epsilon > 0$ there exists an $s \in [0, 1]$ such that
  $f(x^* + \epsilon(x - x^*)) = f(x^*) + \epsilon \nabla f(x^* + s\epsilon(x - x^*))'(x - x^*)$.
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- The above two imply that there exists small $\epsilon > 0$ small such that
  $$f(x^* + \epsilon(x - x^*)) < f(x^*)$$
- The vector $x^* + \epsilon(x - x^*)$ is feasible for all $\epsilon \in [0, 1]$ because $X$ is convex, contradicting the local optimality of $x^*$.

# Proof of Claim b

**Claim (b)** If further assume that, $f$ is a convex function, this condition is also sufficient for $x^*$ to minimize $f$ over $X$

- Using the convexity of $f$

$$f(x) \geq f(x^*) + \nabla f(x^*)'(x - x^*)$$

for every $x \in X$.

- If the condition $\nabla f(x^*)'(x - x^*) \geq 0$ holds for all $x \in X$,we obtain $f(x) \geq f(x^*)$.

- So $x^*$ minimizes $f$ over $X$.

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- For more general cases, or more complex constraints, may not have closed-form solution like above, but provide insight on how to identify optimal solutions

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The following example comes from Book [Example 2.1.1 Version 2]

- Let $X = \{x \mid x \geq 0\}$. Then the necessary condition for $x^* = (x_1^*, \ldots, x_n^*)'$ to be a local min is [recall the previous example]

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- If $x_i^* > 0$ [that is, the $i$th constraint is inactive]. Let $x_j = x_j^*$ for $j \neq i$ and $x_i = \frac{1}{2}x_i^*$. Then $\frac{\partial f(x^*)}{\partial x_i} \leq 0$, so

$$\frac{\partial f(x^*)}{\partial x_i} \doteq 0, \text{ if } x_i^* > 0.$$

[see graph on board]

# Example: Optimization over a Simplex

Let us consider the following simplex constraint:
$X = \{x \mid x \geq 0, \sum_{i=1}^{n} x_i = r\}$, where $r > 0$ is given scalar.

- Necessary condition for $x^* = (x_1^*, \ldots, x_n^*)'$ to be a local min:

  $\sum_{i=1}^{n} \frac{\partial f(x^*)}{\partial x_i}(x_i - x_i^*) \geq 0, \forall x_i \geq 0$ with $\sum_{i=1}^{n} x_i = r$

- Fix $i$ with $x_i^* > 0$ and let $j$ be any other index. Use $x$ with $x_i = 0$, $x_j = x_j^* + x_i^*$, and $x_m = x_m^*$ for all $m \neq i, j$:

$$\left(\frac{\partial f(x^*)}{\partial x_j} - \frac{\partial f(x^*)}{\partial x_i}\right)x_i^* \geq 0,$$

$$x_i^* > 0 \Longrightarrow \frac{\partial f(x^*)}{\partial x_i} \leq \frac{\partial f(x^*)}{\partial x_j}, \forall j.$$

# Projection Over A convex Set

- Projection is a very important operation to deal with constraints [figure]

- **Question:** What if a point is out of the feasible set $X$?

- **Answer:** "Project" it back to $X$!

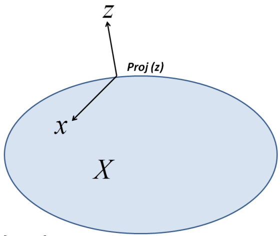


Figure 0.1: Projection over a convex set.

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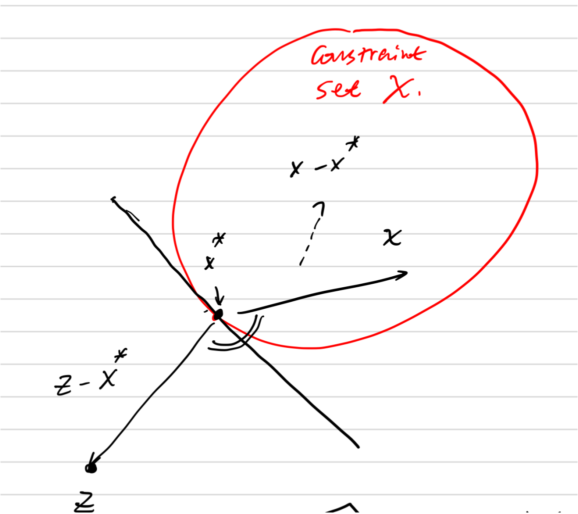
- $x^* = \text{proj}[z] \Longleftrightarrow$ The angle between $z - x^*$ and $x - x^*$ is greater or equal to 90 degrees for all $x \in X$, or $(z - x^*)'(x - x^*) \leq 0$

# Projection Over A convex Set



# Projection Over A convex Set

- **Key Property:** The mapping $f : \mathbb{R}^n \mapsto X$ defined by $f(z) = \text{proj}[z]$ is continuous and satisfying the following property, that is,

$$\|\text{proj}[z] - \text{proj}[y]\| \leq \|z - y\|, \; \forall z, y \in R^{n.}$$

- Intuitively is this true?

- Pic on board

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- **Proof**: From the optimality condition

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- plugin $x = \text{proj}[y]$ in (0.6), and $x = \text{proj}[z]$ in (0.7) (note, these are ok since proj$[y]$ and proj$[z]$ both are feasible, so that are both in $X$ ), and sum:

$$\langle y - \text{proj}[y] - (z - \text{proj}[z]), \text{proj}[z] - \text{proj}[y] \rangle \leq 0, \quad \forall\, x \in X$$

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- move terms around, we obtain

$$\|\mathsf{proj}[z] - \mathsf{proj}[y]\|^2 \leq \langle y - z, \mathsf{proj}[z] - \mathsf{proj}[y] \rangle$$

# Projection Over A convex Set

**Exercise:** Assume $X$ is convex. A vector $x^* \in X$ is a stationary point of

$$\boxed{\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & x \in X, \end{array}}$$

iff $x^*$ satisfies the following fixed point equation

$$x^* = \text{proj}[x^* - \alpha \nabla f(x^*)]$$

for any $\alpha > 0$.